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Perfect Factors in the de Bruijn Graph

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Abstract. A Perfect Factor in the de Bruijn graph is a collection of disjoint cycles of a fixed period with the property that each vertex of the graph lies on exactly one cycle. Alternatively a Perfect Factor is a set of periodic sequences of a fixed period such that for some fixed u, every possible u-tuple of consecutive symbols occurs exactly once as a sub-sequence of a sequence of the set. As well as being of interest in their own right, Perfect Factors are fundamental in the construction of Perfect Maps in two dimensions. In this paper, necessary conditions on the period for the existence of Perfect Factors are given. These conditions are shown to be sufficient in the case where the size of the alphabet is a power of a prime. We also consider the construction of Perfect Factors over general c-ary alphabets. Our results are then applied to obtain a large new class of non-binary Perfect Maps.

1. Introduction

We begin by establishing some notation used throughout this paper.

Let $u \geq 1$ and $c \geq 2$ be positive integers and let $\mathcal{A}$ be any set of cardinality c. We call $\mathcal{A}$ an alphabet of size c. Throughout p is a prime, and α a positive integer. We write $GF(p^\alpha)$ for the (unique) finite field with p^α elements. The elements of $GF(p)$ are represented by the integers modulo p, and regarding $GF(p^\alpha)$ as a vector space of dimension α over $GF(p)$, the element β of $GF(p^\alpha)$ is represented by the vector $(b_0, \ldots, b_{\alpha-1})$ where $b_i \in GF(p)$.

Definition. Following the notation of [2] and generalising from the binary case to the finite field $GF(p^\alpha)$, we let $S(t)$ denote the set of sequences $(s) = \ldots, s_{-1}, s_0, s_1, \ldots$ whose terms are elements of $GF(p^\alpha)$ satisfying $s_i = s_{i+t}$ for every $i \in \mathbf{Z}$. We say that the sequences of $S(t)$ have period t. If t is the least positive integer for which sequence (s) satisfies $s_i = s_{i+t}$, then we say that (s) has *least period* t. We may represent a member (s) of $S(t)$ by its *generating cycle*:

$$(s) = [s_0, s_1, \ldots, s_{t-1}].$$

We can then regard $S(t)$ as a vector space of dimension t over $GF(p^\alpha)$. We say that two sequences in $S(t)$ are *equivalent* if one sequence is just the shift of the other, or, in terms of generating cycles, if the generating cycle of one is just a cyclic shift of the generating cycle of the other. Otherwise they are said to be *inequivalent.*

Definition. The c-ary u^{th} order de Bruijn graph over alphabet $\mathcal{A}$, $G_{u,c}$, is a directed graph with c^u vertices, labelled by the different u-tuples of elements of $\mathcal{A}$. Vertices $x =$

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$(x_1, \ldots, x_u)$ and $y = (y_1, \ldots, y_u)$ of $G_{u,c}$ are joined by an edge from x to y if $(x_2, \ldots, x_u) = (y_1, \ldots, y_{u-1})$. In this case, y is called a successor of x, and each vertex has c successors. Similarly, x is called a predecessor of y, and each vertex has c predecessors.

Since the graph obtained is isomorphic for each choice of alphabet $\mathcal{A}$ of a given size, we will often assume that the elements of $\mathcal{A}$ are the integers $\{0, 1, \ldots, c-1\}$ or the elements of some finite field when $c = p^\alpha$ is a power of a prime; we will then make use of the algebraic properties of the set $\mathcal{A}$.

Definition. A t-cycle in $G_{u,c}$ is simply a directed cycle in $G_{u,c}$ (in the graph theoretical sense) with t distinct vertices.

Clearly a t-cycle in $G_{u,c}$ may be represented by a periodic sequence of elements of $\mathcal{A}$ with generating cycle $[s_0, \ldots, s_{t-1}]$ where $s_i \in \mathcal{A}$ is the first component of the ith vertex of the cycle. With this representation, a *c-ary span u de Bruijn sequence* corresponds to a Hamiltonian cycle in $G_{u,c}$. There are constructions for de Bruijn sequences for arbitrary c and u in [5], [9], [23], [25], [26]. The de Bruijn graph has received much attention in the literature: [9] provides a good survey, while applications have been proposed for the de Bruijn graph and de Bruijn sequences in the fields of communications and coding, [4], [11], [24], position sensing, [1], and in Cryptography, [12].

In the next two definitions, we introduce the principal objects of study in this paper.

Definition. A (u, t, c) Perfect Factor (PF) is a set of vertex disjoint t-cycles in $G_{u,c}$ such that each vertex of $G_{u,c}$ lies on exactly one cycle in the set.

Alternatively, using the correspondence between cycles and sequences, a (u, t, c) Perfect Factor may be thought of as a set of sequences of least period t whose symbols are drawn from an alphabet $\mathcal{A}$ of size c with the property that each of the c^u different u-tuples of $\mathcal{A}^u$ arises once as a sub-sequence in a period of exactly one sequence of the set.

Example 1.1. The pair of sequences:

$$[00, 00, 11, 11, 00, 10, 11, 01], \quad [10, 00, 01, 01, 10, 10, 01, 11]$$

form a $(2, 8, 4)$ Perfect Factor over $GF(2^2)$.

Definition. A c-ary $(r, s; u, v)$ Perfect Map (PM) is an $r \times s$ periodic array with elements from an alphabet $\mathcal{A}$ of size c with the property that each $u \times v$ c-ary array arises once as a sub-array in one period of the array.

We have the following necessary conditions for the existence of a Perfect Factor:

THEOREM 1.2 *Suppose there exists a (u, t, c) PF. Then $t \mid c^u$ and either $u = t = 1$ or $u + 1 \leq t \leq c^u$.*

Proof. The graph $G_{u,c}$ has c^u vertices and in a Perfect Factor the vertex set is partitioned into a number of sets of equal size t. Hence $t \mid c^u$, and so $t \leq c^u$. Supposing $t \leq u$ we

see that each u-tuple consisting of the same element of $\mathcal{A}$ repeated u times either occurs not at all or t times—a contradiction of the properties required of a PF unless $u = t = 1$. □

Similarly to Theorem 1.2 we have:

THEOREM 1.3 *Suppose there exists a c-ary* $(r, s; u, v)$ *Perfect Map. Then*

(i) $rs = c^{uv}$,

(ii) $r = u = 1$ *or* $u + 1 \leq r$,

(iii) $s = v = 1$ *or* $v + 1 \leq s$.

Perfect Factors were first studied in the case $c = 2$ in [15] and the conditions of Theorem 1.2 were shown to be sufficient in that case in [6]. Perfect Factors are of theoretical interest as a property of the de Bruijn graph, but they have also had an important part to play in the construction of other combinatorial objects. In [6], they were used as the building blocks for a class of binary Perfect Maps in two dimensions. In [21] they were again used in this mode, producing a class of Perfect Maps with good 'decoding' properties making them useful in certain position sensing applications. They have also been used in [7] in the construction of a class of de Bruijn sequences having minimal linear complexity.

Likewise, Perfect Maps have been studied mainly in the binary case—In [22], Paterson, building on the results of [6], [8] and [17], showed that the necessary conditions of Theorem 1.3 on r, s, u, v for the existence of a binary $(r, s; u, v)$ Perfect Map are also sufficient. Other authors, notably in [3], [6], [13] and [14], have considered the construction of limited classes of non-binary Perfect Maps.

We point out that a (u, c^u, c) PF is simply a de Bruijn cycle in $G_{u,c}$. Note also that when $u = 1$ the necessary conditions of Theorem 1.2 are trivially sufficient: to obtain our cycles we simply partition the set $\mathcal{A}$ into $\frac{c}{t}$ ordered sets each of size t. Consequently, in the remainder of this paper we shall assume that $u \geq 2$.

When c is a power of a prime, $c = p^\alpha$, the conditions of Theorem 1.2 become:

$$t = p^k \quad \text{and} \quad u + 1 \leq p^k \leq p^{\alpha u}.$$

In Theorem 4.6 we show that this condition is actually sufficient when c is a power of a prime. We build up to this result in Section 2, by studying properties of the linear complexity of sequences over a finite field, and in Section 3 by considering the case where $c = p$, a prime.

Our results in Sections 5 and 6 concern the existence of Perfect Factors and Perfect Maps over general c-ary alphabets. We construct a large class of such Perfect Factors by combining PFs over alphabets having sizes that are powers of different primes in a simple way in Section 5. There we also indicate how far our constructions go in proving the sufficiency of Theorem 1.2 for general c. We apply these PFs in a generalised version of the constructions of [6] and [21] in Section 6. We obtain a large new class of non-binary Perfect Maps similar to the those first found in [6] in the binary case.

2. Linear Complexity and a Class of Perfect Factors

We begin with a study of the linear complexity of sequences over $GF(p^\alpha)$ having special periods. The results of this study are used to construct a new set of PFs using linear complexity as a tool.

Definition. Suppose that the sequence $(s) = \ldots s_{-1}, s_0, s_1, \ldots$ of terms of $GF(p^\alpha)$ satisfies:

$$s_{i+u} = c_0 s_i + c_1 s_{i+1} + \cdots + c_{u-1} s_{i+u-1} \quad \text{for all } i \in \mathbf{Z},$$

that is, *a linear recurrence relation of degree* u. Then the polynomial $g(X)$ given by

$$g(X) = c_0 + c_1 X + c_2 X^2 + \cdots + c_{u-1} X^{u-1} + X^u$$

is called the *characteristic polynomial* of the corresponding linear recurrence.

It is clear that any periodic sequence satisfies some linear recurrence, for if $(s) \in S(t)$, then (s) satisfies

$$s_{i+t} = s_i \quad \text{for all } i \in \mathbf{Z},$$

a linear recurrence of degree t. We have the following Result and Definitions.

Result 2.1. [16], Theorem 6.42.

Let $(s) \in S(t)$. Then there exists a uniquely determined monic polynomial $m(X) \in GF(p^\alpha)[X]$ having the following property: a polynomial $g(X) \in GF(p^\alpha)[X]$ is the characteristic polynomial of a linear recurrence generating (s) if and only if $m(X)$ divides $g(X)$.

Definition. The polynomial $m(X)$ in Result 2.1 is called the *minimal polynomial* of (s).

Definition. Suppose $(s) \in S(t)$. Then the *linear complexity* of (s) is the least positive integer c such that (s) satisfies a linear recurrence of degree c. We frequently write $c(s)$ for the linear complexity of the sequence (s).

From the above discussion we see that the linear complexity of a periodic sequence (s) is the degree of the minimal polynomial $m(X)$ of (s).

Definition. Let $(s) = \ldots s_{-1}, s_0, s_1, \ldots$ be an arbitrary sequence over $GF(p^\alpha)$. We define the *left shift operator* E acting on the sequence (s) by defining $E(s)$ to be the sequence whose ith term is s_{i+1}.

If $m(X)$ is the minimal polynomial of (s), we have, for all $i \in \mathbf{Z}$,

$$m(E)(s) = [0],$$

the all-zero sequence. When $(s) \in S(t)$, $(E^t - 1)(s) = [0]$ and so $m(E) \mid E^t - 1$. Further when $t = p^k$, we have $E^t - 1 = (E-1)^t$, since we are working in a field of characteristic p. Then the minimal polynomial of (s) is $m(E) = (E-1)^{c(s)}$.

Following [10] for the binary case, we examine the binomial coefficients over $GF(p^\alpha)$ and their relationship to the linear complexities of sequences over $GF(p^\alpha)$.

For non-negative integers i and j, the binomial coefficient $\binom{i}{j}$ is defined over the non-negative integers in the usual way, with the convention that $\binom{i}{j} = 0$ when $i < j$. The binomial coefficients $\binom{i}{j}$ are defined over $GF(p^\alpha)$ by interpreting the coefficient over the integers modulo p and regarding $GF(p)$ as a subfield of $GF(p^\alpha)$. For j fixed and non-negative, we define a semi-infinite sequence $(b)^j$ by

$$(b)^j = \binom{0}{j}, \binom{1}{j}, \binom{2}{j}, \ldots.$$

There is a special relationship between the sequences $(b)^j$ and the linear complexity of sequences of $S(p^k)$ which we investigate in the next few results.

LEMMA 2.2 *The sequence $(b)^j$ has linear complexity $j+1$ and least period p^k where k is the least integer such that $j < p^k$.*

Proof. Consider the binomial coefficient identity

$$\binom{i+1}{j} = \binom{i}{j} + \binom{i}{j-1}.$$

Re-arranging and using the operator E we have:

$$(E-1)(b)^j = (b)^{j-1}$$

and so for every integer $m \leq j$,

$$(E-1)^m (b)^j = (b)^{j-m}.$$

In particular, if k is the least integer such that $j < p^k$, then

$$(E^{p^{k-1}} - 1)(b)^j = (E-1)^{p^{k-1}}(b)^j = (b)^{j-p^{k-1}}$$

which is not the all-zero sequence, while

$$(E^{p^k} - 1)(b)^j = (E-1)^{p^k}(b)^j = (E-1)^{p^k-j-1}(E-1)^{j+1}(b)^j = [0].$$

Thus $(b)^j$ has least period p^k. Further,

$$(E-1)^j(b)^j = (b)^0 = [1] \quad \text{and} \quad (E-1)^{j+1}(b)^j = [0].$$

The result follows from the discussion above on the minimal polynomials of sequences of $S(p^k)$. □

As a consequence of Lemma 2.2, we may regard the sequences $(b)^j$ as being infinite sequences of periods p^k. We have the following Definition and Lemma.

Definition. For non-negative integers k, we define

$$\mathcal{B}_k = \{(b)^0, (b)^1, \ldots, (b)^{p^k-1}\}.$$

LEMMA 2.3 *For each positive integer k, the sequences of $\mathcal{B}_k$ form a basis for the vector space $S(p^k)$, called the binomial basis.*

Proof. The p^k distinct sequences of $\mathcal{B}_k$ are in $S(p^k)$, a space of dimension p^k. Moreover, since $\binom{i}{i} = 1$ and $\binom{i}{j} = 0$ for $i < j$, the sequences of $\mathcal{B}_k$ are linearly independent. □

THEOREM 2.4 *Let $(s) \in S(p^k)$. Then (s) has linear complexity c if and only if, writing (s) uniquely as a linear combination of the vectors of basis $\mathcal{B}_k$,*

$$(s) = \sum_{j=0}^{p^k-1} a_j (b)^j,$$

we have:

$$\begin{aligned} a_{c-1} &\neq 0, \\ a_j &= 0 \quad \text{for } j \geq c. \end{aligned}$$

Proof. Suppose $(s) \in S(p^k)$ has linear complexity c. Writing

$$(s) = \sum_{j=0}^{p^k-1} a_j (b)^j,$$

the proof of the forward implication quickly follows: note that

$$\begin{aligned} [0] = (E-1)^c (s) &= (E-1)^c \sum_{j=0}^{p^k-1} a_j (b)^j \\ &= \sum_{j=0}^{p^k-1} a_j (E-1)^c (b)^j \\ &= \sum_{j=c}^{p^k-1} a_j (b)^{j-c} \quad \text{since } (E-1)^c (b)^j = [0] \text{ for } j < c \end{aligned}$$

while the sequences $(b)^j$ are linearly independent, so we must have $a_c = a_{c+1} = \cdots = a_{p^k-1} = 0$. But $(E-1)^{c-1}(s)$ is not the zero sequence, and

$$\begin{aligned} (E-1)^{c-1}(s) &= (E-1)^{c-1} \sum_{j=0}^{c-1} a_j (b)^j \\ &= [a_{c-1}] \end{aligned}$$

so that a_{c-1} is non-zero.

The converse follows quickly from the observation that the sequence $(b)^j$ has linear complexity $j+1$. □

Theorem 2.4 is essentially Theorem 2 of [10] generalised to an arbitrary finite field. Notice that contained in the proof of Theorem 2.4 we have the following characterisation, identical in the binary case to that of [2].

THEOREM 2.5 *$(s) \in S(p^k)$ has linear complexity c if and only if there exists a non-zero $a_{c-1} \in GF(p^\alpha)$ such that*

$$(E-1)^{c-1}(s) = [a_{c-1}].$$

The import of Theorem 2.4 is that, if we view sequences relative to the basis $\mathcal{B}_k$, then we can immediately recover information on their linear complexity.

We define the *weight* of a sequence $(s) \in S(t)$, denoted by $\mathrm{wt}(s)$, to be the sum $s_0 + s_1 + \cdots + s_{t-1}$. Note that the period t here need not be the least possible period of (s).

COROLLARY 2.6 *Suppose (s) has least period p^k and linear complexity c. Then*

$$p^{k-1} + 1 \leq c \leq p^k$$

and $c = p^k$ if and only if (s) has non-zero weight.

Proof. $(s) \in S(p^k)$, so we may write $(s) = \sum_{j=0}^{p^k-1} a_j (b)^j$. By Theorem 2.4 we immediately have that $c \leq p^k$.

Similarly, if $c \leq p^{k-1}$, then we can write $(s) = \sum_{j=0}^{p^{k-1}-1} a_j (b)^j$ and, from the periodicity of the $(b)^j$, we see that (s) has period p^{k-1}, a contradiction. So $c > p^{k-1}$.

Now $c = p^k$ if and only if there exists $a_{c-1} \neq 0$ such that $(E-1)^{p^k-1}(s) = [a_{c-1}]$ by Theorem 2.5. But

$$(E-1)^{p^k-1} = \frac{(E^{p^k}-1)}{(E-1)} = E^{p^k-1} + E^{p^k-2} + \cdots + E + 1$$

and so $(E-1)^{p^k-1}(s) = [a_{c-1}]$ is equivalent to writing

$$s_{i+p^k-1} + s_{i+p^k-2} + \cdots + s_{i+1} + s_i = a_{c-1},$$

i.e. equivalent to $wt(s) = a_{c-1}$. The second part of the Corollary follows. □

By the discussion above on characteristic polynomials of sequences of period p^k, we see that a sequence in $S(p^k)$ has minimal polynomial of the form $m(E) = (E-1)^c$ for some c, so that the coefficients of $m(E)$ are elements of $GF(p)$. Then, in calculating linear complexities of sequences of period p^k, we may work component-wise: that is if $(s) = \ldots s_{-1}, s_0, s_1, \ldots$ and $s_i = (s_i^1, \ldots, s_i^\alpha)$ where $s_i^j \in GF(p)$, then we have α

component sequences over $GF(p)$,

$$(s)^j = \ldots s_{-1}^j, s_0^j, s_1^j, \ldots,$$

each of period p^k, with

$$(s) = ((s)^1, \ldots, (s)^\alpha).$$

In the notation of Theorem 2.4 and Theorem 2.5,

$$(E-1)^{c-1}(s) = [a], \quad \text{for some } a \neq 0$$

if and only if

$$(E-1)^{c-1}(s)^j = [a_j] \quad \text{for some } a = (a_1, \ldots, a_\alpha) \neq (0, \ldots, 0).$$

Then it is clear that

$$c(s) = \max\{c((s)^j) : 1 \leq j \leq \alpha\}.$$

We make use of this fact in subsequent Sections.

We can use the properties of the binomial basis to construct a class of Perfect Factors.

THEOREM 2.7 *Suppose $p^{k-1} + 1 \leq c \leq p^k$ and non-zero $a \in GF(p^\alpha)$ are fixed. Let $\mathcal{F}$ be the set of sequences (s) of period p^k satisfying*

$$(E-1)^{c-1}(s) = [a].$$

Then $\mathcal{F}$ gives rise to a set of inequivalent sequences $\mathcal{G}$ of size $p^{\alpha(c-1)-k}$ with the property that every $(c-1)$-tuple of elements of $GF(p^\alpha)$ occurs exactly once as a sub-sequence of one sequence of $\mathcal{G}$. Moreover the sequences of $\mathcal{G}$ all have the same linear complexity c.

Proof. Theorem 2.4 shows that the set $\mathcal{F}$ of sequences $(s) \in S(p^k)$ satisfying $(E-1)^{c-1}(s) = [a]$ are precisely those with $a_{c-1} = a$ and $a_c = a_{c+1} = \cdots = a_{p^k-1} = 0$, where $(s) = \sum_{i=0}^{p^k-1} a_j(b)^j$. By the bounds on c and Theorem 2.4, the sequences of $\mathcal{F}$ have least period p^k and linear complexity c.

Since $\mathcal{B}_k$ is a basis for $S(p^k)$, distinct choices for the a_j's lead to distinct sequences. There remain $p^{\alpha(c-1)}$ choices for $a_0, \ldots, a_{c-2}$ and so the set of sequences $\mathcal{F}$ is of size $p^{\alpha(c-1)}$. Now let $(s) \in \mathcal{F}$. Then (s) satisfies $(E-1)^{c-1}(s) = [a]$ and any sequence that is a shift of (s) also satisfies the same linear relation. We deduce that each $(s) \in \mathcal{F}$ gives rise to p^k equivalent sequences also contained in $\mathcal{F}$, namely the p^k distinct shifts of (s). Hence $\mathcal{F}$ gives a set of inequivalent sequences of size $p^{\alpha(c-1)-k}$, which we denote by $\mathcal{G}$.

Between them, the sequences of $\mathcal{G}$ contain at most $p^{\alpha(c-1)}$ distinct $(c-1)$ sub-sequences. Therefore to show that each $(c-1)$-tuple of elements of $GF(p^\alpha)$ occurs exactly once as a sub-sequence of a sequence of $\mathcal{G}$, it is enough to show that an arbitrary $(c-1)$-tuple $(d_0, \ldots, d_{c-2})$ occurs as a sub-sequence in some sequence of $\mathcal{F}$.

Let

$$\begin{aligned} a_0 &= d_0, \\ a_j &= d_j - \sum_{k=0}^{j-1} a_k \binom{j}{k} \\ a_{c-1} &= a \\ a_j &= 0 \quad (c \le j \le p^k - 1) \end{aligned}$$

and

$$(s) = \sum_{j=0}^{p^k-1} a_j (b)^j.$$

We claim that $(s) \in \mathcal{F}$ has $(d_0, \ldots, d_{c-2})$ as its first $c-1$ terms $s_0, \ldots, s_{c-2}$. Firstly,

$$s_0 = \sum_{j=0}^{p^k-1} a_j \binom{0}{j} = a_0 = d_0$$

and for $1 \le l \le c-2$,

$$\begin{aligned} s_l &= \sum_{j=0}^{p^k-1} a_j \binom{l}{j} \\ &= \sum_{j=0}^{l} a_j \binom{l}{j} + 0 \quad \text{since } \binom{l}{j} = 0 \text{ for } j > l \\ &= a_l + \sum_{j=0}^{l-1} a_j \binom{l}{j} \\ &= d_l - \sum_{j=0}^{l-1} a_j \binom{l}{j} + \sum_{j=0}^{l-1} a_j \binom{l}{j} \\ &= d_l. \end{aligned}$$

The result follows. □

As a consequence of Theorem 2.7 we have:

COROLLARY 2.8 *Let p^α be a power of a prime. Then for any $k \ge 1$ and any u with $p^{k-1} \le u < p^k$, there exists a (u, p^k, p^α) Perfect Factor consisting of sequences of complexity $u+1$ over $GF(p^\alpha)$.*

Proof. From Theorem 2.7, we deduce that for each $0 \ne a \in GF(p^\alpha)$, the inequivalent sequences (s) of period p^k satisfying $(E-1)^u(s) = [a]$ constitute a set of cycles of complexity $u+1$ forming a Perfect Factor with the appropriate parameters. Indeed we obtain $p^\alpha - 1$ different Perfect Factors with the same parameters, one for each choice of a: fixing on a choice of a gives us one PF and we can obtain all the other PFs from the

sequences of that PF by simply considering the scalar multiples of the factor sequences by each non-zero $\lambda \in GF(p^\alpha)$ in turn. □

Example 2.9. We construct a $(3, 9, 3)$ PF over $GF(3)$; according to Corollary 2.8, the sequences of such a Perfect Factor can be obtained as the set of inequivalent sequences obtained by taking linear combinations of the binomial sequences

$$\begin{aligned}(b)^0 &= [1,1,1,1,1,1,1,1,1],\\ (b)^1 &= [0,1,2,0,1,2,0,1,2],\\ (b)^2 &= [0,0,1,0,0,1,0,0,1],\\ (b)^3 &= [0,0,0,1,1,1,2,2,2],\end{aligned}$$

with a fixed coefficient a for $(b)^3$. We take $a = 1$ to obtain the $(3, 9, 3)$ PF:

$$[0,0,0,1,1,1,2,2,2],[0,1,2,1,2,0,2,0,1],[0,0,2,1,1,0,2,2,1].$$

3. Lempel's Homomorphism and Perfect Factors over $GF(p)$

In this section we introduce a generalisation of Lempel's homomorphism, [15], for the de Bruijn graphs G_{u,p^α}. We then use properties of this map and the results of Section 2 to construct PFs for all the parameter sets of Theorem 1.2 in the case where $c = p$, a prime.

Let $GF(p^\alpha)^u$ denote the set of u-tuples of elements of $GF(p^\alpha)$, so the elements of $GF(p^\alpha)^u$ can be regarded as the vertices of G_{u,p^α}. Define a mapping $D : GF(p^\alpha)^u \to GF(p^\alpha)^{u-1}$ by

$$D(x_1, \ldots, x_u) = (x_2 - x_1, \ldots, x_u - x_{u-1})$$

LEMMA 3.1 *D is a graph homomorphism from G_{u,p^α} onto G_{u-1,p^α}. For all $x, y \in GF(p^\alpha)^u$, $Dx = Dy$ if and only if $x = y + (a, \ldots, a)$ for some $a \in GF(p^\alpha)$. D is the generalisation of Lempel's homomorphism [15] to the field $GF(p^\alpha)$.*

Proof. The proof is analogous to that of [15], Theorem 4 and Lemma 2. □

LEMMA 3.2 *Given $x = (x_1, \ldots, x_{u-1}) \in GF(p^\alpha)^{u-1}$, x has p^α pre-images under D^{-1}, namely*

$$\left\{\left(a, a + x_1, a + x_1 + x_2, \ldots, a + \sum_{i=1}^{u-1} x_i\right) : a \in GF(p^\alpha)\right\}.$$

Proof. The proof is analogous to that of [15], Lemma 1. □

Since D is a graph homomorphism, we may apply D to a cycle of vertices $v_1, \ldots, v_t$ in G_{u,p^α} to obtain a sequence of (not necessarily distinct) vertices $Dv_1, \ldots, Dv_t$ in G_{u-1,p^α} with the property that there is a directed edge from Dv_i to Dv_{i+1} $(1 \leq i \leq t-1)$ and from Dv_t to Dv_1. If the cycle in G_{u,p^α} corresponds to a sequence with generating cycle $[s_1, \ldots, s_t]$ then the resulting sequence of vertices in G_{u-1,p^α} corresponds to a cycle $[s_2 - s_1, s_3 - s_2, \ldots, s_t - s_{t-1}, s_1 - s_t]$. Thus we see that applying D to a cycle in G_{u,p^α} is the same as applying the operator $E - 1$ to the corresponding generating cycle. However the operator $E - 1$ is more general, as we have only defined D acting on sequences arising from cycles in the de Bruijn graph.

Given a cycle of vertices in G_{u,p^α} we can also consider the pre-images of the cycle under D^{-1}.

We have two theorems based on [15]:

THEOREM 3.3 *Suppose* $(s) \in S(t)$ *has zero weight, generating cycle* $[s_0, \ldots, s_{t-1}]$ *and arises from a cycle in* G_{u,p^α}. *Then the set of pre-images of* (s) *under* D^{-1}, *denoted by* $D^{-1}(s)$, *consists of* p^α *inequivalent sequences of period* t *having generating cycles given by*

$$D^{-1}(s) = \left\{ \left[a, a + s_0, a + s_0 + s_1, \ldots, a + \sum_{i=0}^{t-2} s_i \right] : a \in GF(p^\alpha) \right\}.$$

Proof. The proof is analogous to that of [15], Corollary 1. □

THEOREM 3.4 *Suppose* $(s) \in S(t)$ *has non-zero weight, generating cycle* $[s_0, \ldots, s_{t-1}]$ *and arises from a cycle in* G_{u,p^α}. *Then* $D^{-1}(s)$ *consists of* $p^{\alpha-1}$ *inequivalent sequences of period* pt *with generating cycles given by*

$$\begin{aligned} D^{-1}(s) = \Bigg\{ \Bigg(\Bigg[& a, a + s_0, a + s_0 + s_1, \ldots, a + \sum_{i=0}^{t-2} s_i, \\ & a + wt(s), a + wt(s) + s_0, \ldots, a + wt(s) + \sum_{i=0}^{t-2} s_i, \\ & a + 2wt(s), a + 2wt(s) + s_0, \ldots, a + 2wt(s) + \sum_{i=0}^{t-2} s_i, \\ & \vdots \\ & a + (p-1)wt(s), a + (p-1)wt(s) + s_0, \ldots, \\ & a + (p-1)wt(s) + \sum_{i=0}^{t-2} s_i \Bigg] : a \in GF(p^\alpha) \Bigg\}. \end{aligned}$$

Proof. The proof is analogous to that of [15], Corollary 2. □

Note that, here, the sequences whose generating cycles start with the terms a, $a+\mathrm{wt}(s)$, ... and $a+(p-1)\mathrm{wt}(s)$ are all equivalent.

From the fact that applying D is equivalent to using the operator $E-1$ combined with Theorem 2.5, we deduce that when t is a power of p, say $t=p^k$, an application of D to a sequence (s) arising from a t-cycle in G_{u,p^α} produces a sequence having linear complexity one less than that of (s), and conversely the sequences in $D^{-1}(s)$ have linear complexity one greater than that of (s). When the weight of (s) is zero and the linear complexity of (s) is c $(c<p^k)$ we can apply Theorem 3.3 to obtain a set of p^α sequences of period p^k and linear complexity $c+1\leq p^k$. When the weight of (s) is non-zero (and so the linear complexity of (s) is $c=p^k$), by Theorem 3.4 we can obtain a set of $p^{\alpha-1}$ sequences of period p^{k+1} and linear complexity $c+1=p^k+1$. These observations can be used to generate the Perfect Factors of Theorem 2.7—starting with the sequence $[a]$, where $0\neq a\in GF(p^\alpha)$, we apply D^{-1} a total of $c-1$ times to obtain a set of sequences of complexity c. A little thought shows this set to be the same as the set $\mathcal{F}$ obtained in Theorem 2.7. However we prefer to use the basis approach of the previous Section as it reveals more naturally the structure of sequences of a given linear complexity.

Shortly, we use these properties of Lempel's homomorphism in a construction for another class of Perfect Factors; firstly we need to study the linear complexities of a special set of de Bruijn sequences. We adapt a result of [2]:

LEMMA 3.5 *Let (s) be a maximal length recurring sequence (m-sequence) of period $p^{\alpha u}-1$ over $GF(p^\alpha)$ generated by a linear recurrence with characteristic polynomial of degree u, primitive over $GF(p^\alpha)$, and suppose that the generating cycle for (s) begins with $u-1$ zeros. Let (t) denote the span u de Bruijn sequence obtained from (s) by inserting an extra zero at the start of the generating cycle for (s). Then (t) has linear complexity $p^{\alpha u}-1$, unless $p^\alpha=2$ and $u=1$, when (t) has linear complexity 2.*

Proof. The proof of this result follows exactly that of [2], Corollary 6, using the fact that the result of [18] used in the proof there is true over any field. □

THEOREM 3.6 *Suppose p is prime and $u\geq 2$ with $u+1\leq p^k\leq p^u$. Then there exists a (u, p^k, p) Perfect Factor.*

Proof. When $u+1\leq p^k\leq pu$, Corollary 2.8 in the case $\alpha=1$ provides a Perfect Factor with the correct parameters. Suppose then that $pu<p^k\leq p^u$.

When $p^k=p^u$, a Perfect Factor consists of a single de Bruijn sequence of span u over an alphabet of size p.

Otherwise, $pu<p^k\leq p^{u-1}$ where we may assume that $k\geq 2$ (since $p^k>pu>p$). We can similarly rule out the possibility that $p=2$ and $k=2$. From Lemma 3.5, there exists a span $k-1$ de Bruijn sequence (t) over $GF(p)$ of linear complexity $p^{k-1}-1$. From Theorem 3.4, applying D^{-1} to (t) gives a set of p sequences of period p^{k-1} and linear complexity p^{k-1}. Since (t) corresponded to a Hamiltonian cycle in $G_{k-1,p}$, it is easily

shown (using Lemma 3.2) that the resulting sequences correspond to cycles in $G_{k,p}$ and that every vertex of $G_{k,p}$ lies on one of these cycles.

Applying D^{-1} again to each of the set of p sequences, Theorem 3.3 shows that we obtain a new set of p sequences of period p^k and linear complexity $p^{k-1}+1$. Again, it is easily shown that the resulting sequences correspond to cycles in $G_{k+1,p}$ and that every vertex of $G_{k+1,p}$ lies on one of the cycles.

We now apply D^{-1} a further $u-(k+1)$ times. Note that $k \leq u-1$ so that $u-(k+1) \geq 0$ and $pu \leq p^k$ so $u \leq p^{k-1}$ and then $p^{k-1}+u-k \leq p^k$. Similar reasoning to that above then shows that we obtain a set of p^{u-k} sequences of period p^k and complexity $p^{k-1}+u-k$ corresponding with cycles in $G_{u,p}$ having the property that each vertex of $G_{u,p}$ lies on exactly one of the cycles. This set is nothing other than a (u, p^k, p) PF.

Thus we have constructed a (u, p^k, p) PF for every parameter set in the statement of the theorem. □

4. Perfect Factors over a Finite Field

Given prime p and $\alpha \geq 1$, we aim to construct (u, p^k, p^α) PFs for every k subject to the conditions of Theorem 1.2. We initially give some basic construction methods to produce new PFs from other PFs and from PMs. We then combine these methods to give an inductive (but constructive) proof of our result. We continue to use linear complexity to ensure control over the periods of our sequences.

Firstly we give a simple construction which allows us to obtain a PF over an alphabet of size c^α from a PF over an alphabet of size c.

LEMMA 4.1 *Let Γ be a (u, t, c) PF over alphabet $\mathcal{A}$ and let $\alpha \geq 1$. For each choice of α (not necessarily distinct) sequences $(s)^1, \ldots, (s)^\alpha$ from Γ and each choice of shifts $r_2, \ldots, r_\alpha$ with $0 \leq r_i \leq t-1$, we define a c^α-ary sequence (w) over alphabet $\mathcal{A}^\alpha$ as follows: $(w) = [w_0, \ldots, w_{t-1}]$ where*

$$w_i = (s_i^1, s_{i+r_2}^2, \ldots, s_{i+r_\alpha}^\alpha)$$

so that the ith component sequence of (w) is the shift by r_i of $(s)^i$. Then the set of sequences (w), denoted by Γ', form a (u, t, c^α) PF over alphabet $\mathcal{A}^\alpha$.

Proof. Given any u-tuple of elements of $\mathcal{A}^\alpha$ it is easily seen that there exists a choice of sequences from Γ and a choice of shifts such that the tuple occurs as a sub-sequence of a sequence $(w) \in \Gamma'$. □

We note that if $c = p$ and the alphabet $\mathcal{A}$ of Lemma 4.1 is $GF(p)$ then we can use the arguments on the linear complexity of component sequences in Section 2 above. We see that the sequences of the new PF, Γ', in Lemma 4.1 can be regarded as having component sequences over $GF(p)$ from Γ and so the sequences of Γ' have linear complexities bounded above by the maximal linear complexity of the sequences of Γ. We have:

COROLLARY 4.2 *Let Γ be a (u, p^k, p) PF over $GF(p)$ consisting of sequences $(s)^i$, $1 \leq i \leq p^{u-k}$ and define $c(\Gamma) = max\{c((s)^i) : 1 \leq i \leq p^{u-k}\}$. For every $\alpha \geq 1$, there exists a (u, p^k, p^α) PF Γ' over $GF(p^\alpha)$ whose sequences all have linear complexities less than or equal to $c(\Gamma)$.*

Our next construction produces PFs from PMs:

LEMMA 4.3 *Suppose there exists a c-ary $(r, s; u, v)$ Perfect Map over alphabet $\mathcal{A}$. Then there exists a (u, r, c^v) PF over alphabet $\mathcal{A}^v$.*

Proof. Let A be a c-ary $(r, s; u, v)$ Perfect Map over alphabet $\mathcal{A}$ of size c. Let $(c)^j$ denote the jth column of A, a sequence of period r. For each i with $0 \leq i < s$, we define a sequence $(w)^i$ over $\mathcal{A}^v$ by defining $(w)^i$ to have component sequences $((c)^i, (c)^{i+1}, \ldots, (c)^{i+v-1})$, a set of v consecutive columns of A. The properties of A imply that the set of sequences $\{(w)^i : 0 \leq i < s\}$, form a PF with the required parameters. □

We give another construction for new PFs from old that is a generalisation of the construction for de Bruijn sequences of [21], Theorem 5.2.

LEMMA 4.4 *Let p be a prime and suppose there exists a $(u, p^{k-\beta u}, p^{\alpha-\beta})$ PF, Γ, over $GF(p^{\alpha-\beta})$. Then there exists a (u, p^k, p^α) PF Γ' over $GF(p^\alpha)$ whose sequences have linear complexities c with*

$$c \leq p^k - p^{k-\beta u}.$$

Proof. Let (s) be a p^β-ary span u de Bruijn sequence over $GF(p^\beta)$ beginning with u zeros and let $(s)'$ denote the "punctured de Bruijn" sequence derived from (s) by deleting one zero from the block of u zeros. Then $(s)'$ has period $p^{\beta u} - 1$.

For each sequence (t) of Γ, we construct a sequence (w) over $GF(p)^\alpha$ having period p^k as follows. Such a sequence may be regarded as having α component sequences over $GF(p)$: we take the first $\alpha - \beta$ component sequences of (w) to be the components over $GF(p)$ of $p^{\beta u}$ repetitions of (t), and the last β component sequences of (w) to be the components over $GF(p)$ of a run of $p^{k-\beta u}$ terms $(0, \ldots, 0) \in GF(p^\beta)$ followed by $p^{k-\beta u}$ copies of $(s)'$.

We claim that the set of sequences (w) obtained in this way, denoted by Γ', is a (u, p^k, p^α) Perfect Factor over $GF(p^\alpha)$.

Let $\mathbf{x} = (x_1, \ldots, x_u)$ be an arbitrary u-tuple of elements of $GF(p^\alpha)$. We regard the components of each x_i as falling into two groups: the first $\alpha - \beta$ components, y_i, and the last β components z_i. Let $\mathbf{y} = (y_1, \ldots, y_u)$ and $\mathbf{z} = (z_1, \ldots, z_u)$. Then from the properties of Γ, there exists a sequence $(t) \in \Gamma$ such that $\mathbf{y} = (t_i, t_{i+1}, \ldots, t_{i+u-1})$ for some $0 \leq i < p^{k-\beta u}$.

If $\mathbf{z}$ is identically zero, then it is clear that the sequence $(w) \in \Gamma'$ corresponding to (t) satisfies $\mathbf{x} = (w_i, w_{i+1}, \ldots, w_{i+u-1})$. Otherwise, $\mathbf{z}$ is not zero and from the properties of $(s)'$

we have $\mathbf{z} = (s'_j, s'_{j+1}, \ldots, s'_{j+u-1})$ for some $0 \le j < p^{\beta u} - 1$. Since $\gcd(p^{k-\beta u}, p^{\beta u} - 1) = 1$, there exists a unique integer r with $p^{k-\beta u} \le r < p^k$ such that:

$$r \equiv i \pmod{p^{k-\beta u}}$$
$$r \equiv j + p^{k-\beta u} \pmod{p^{\beta u} - 1}.$$

Then from the construction of (w) we have that $\mathbf{x}$ occurs as the sub-sequence $(w_r, w_{r+1}, \ldots, w_{r+u-1})$ of (w). Thus we have shown that an arbitrary u-tuple $\mathbf{x}$ occurs as a sub-sequence in some sequence of Γ'.

We concentrate now on the linear complexity of our Factor sequences. It is enough to prove that their component sequences over $GF(p)$ have linear complexities satisfying the bound of the Theorem, in view of the discussion above on the relationship between linear complexities of sequences and their component sequences.

Examining the first $\alpha - \beta$ component sequences of the sequences of Γ', we see that they are formed from sequences of period $p^{k-\beta u}$ over $GF(p)$ and so they have complexities $c \le p^{k-\beta u}$. Notice that since $p \ge 2$ and $\beta u \ge 2$, $p^{k-\beta u} \le p^k - p^{k-\beta u}$.

Fixing on a sequence $(t) \in \Gamma$, we consider the final β components of the corresponding (w) as a single sequence $(a) = [a_0, \ldots, a_{p^k-1}]$ over $GF(p^\beta)$ of period p^k. For any $j \in \mathbf{Z}$, consider

$$a_j + a_{j+m} + a_{j+2m} + \cdots + a_{j+(p^\beta-1)m} \tag{1}$$

where $m = p^{k-\beta u}$. Since $(s)'$ has period $p^{\beta u} - 1$ and $\gcd(p^{\beta u} - 1, m) = 1$, in equation (1) we pick out each term of $(s)'$ exactly once in some order, along with one zero from the block of m zeros. But (s) is a de Bruijn sequence of span u and so

$$\begin{aligned} \sum_{i=0}^{p^{\beta u}-1} a_{j+im} &= \sum_{i=0}^{p^{\beta u}-1} s_i \\ &= (p^\beta)^{u-1} \sum_{x \in GF(p^\beta)} x \\ &= 0 \end{aligned}$$

since $\beta(u-1) \ge 1$ and $GF(p^\beta)$ has characteristic p. Hence the sequence (a) arising from the last β components of (w) satisfies a linear recurrence of degree $(p^{\beta u} - 1)p^{k-\beta u} = p^k - p^{k-\beta u}$ and then so must the component sequences over $GF(p)$ that make up (a). So the last β components of (w) individually satisfy this same recurrence. Thus they have linear complexities less than or equal to $p^k - p^{k-\beta u}$.

We deduce that the component sequences of (w) each satisfy linear recurrences of degrees less than or equal $p^k - p^{k-\beta u}$ and the result follows. □

Our final construction for PFs imitates our previous use of D^{-1} to construct PFs in Theorem 3.6.

LEMMA 4.5 *Suppose there exists a (u, p^k, p^α) PF, Γ over $GF(p^\alpha)$ whose sequences all have linear complexities less than or equal to c, where $c < p^k$. Then the set of pre-images*

of the sequences of Γ *under* D^{-1} *forms a* $(u+1, p^k, p^\alpha)$ *PF,* Γ'*, over* $GF(p^\alpha)$ *whose sequences all have linear complexities less than or equal to* $c+1$.

Proof. Let Γ' denote the set of pre-images of the sequences of Γ under D^{-1}. The sequences of Γ have period p^k and linear complexity less than or equal $c < p^k$. Our previous discussion implies that the sequences of Γ' also have period p^k and linear complexity less than or equal $c+1$. Since Γ corresponds to a set of cycles in G_{u,p^α} with the property that every vertex of G_{u,p^α} lies on one of the cycles, it is easily shown (using Lemma 3.2) that the resulting sequences in Γ' correspond to cycles in G_{u+1,p^α} and that every vertex of G_{u+1,p^α} lies on one of these cycles. The result follows. □

We now combine the preceding Lemmas to produce our first major result.

THEOREM 4.6 *Suppose* p *is a prime,* $\alpha \geq 1$ *and* $u \geq 1$. *Suppose further that*

$$u+1 \leq p^k \leq p^{\alpha u}.$$

Then there exists a (u, p^k, p^α) *Perfect Factor. That is, the conditions of Theorem 1.2 are sufficient for the existence of a Perfect Factor when* c *is a power of a prime.*

Proof. In the case $u = 1$, we have already noted the sufficiency of the conditions of Theorem 1.2. We therefore assume that $u \geq 2$. We establish Theorem 4.6 by induction on α.

When $\alpha = 1$, the result is true by Theorem 3.6. Suppose then that $\alpha \geq 2$ and the result is true for alphabets of size $p^{\alpha-1}$.

We first consider the cases where $u+1 \leq p^k \leq p^u$. For these values of k, there exists a (u, p^k, p) PF by Theorem 3.6. Applying Corollary 4.2 we obtain a (u, p^k, p^α) PF.

Secondly we consider the cases where

$$p^{u+1} \leq p^k < (u+1)p^u. \tag{2}$$

Notice that for $\alpha \geq 2$, $u \geq 2$ and p prime, $u+1 < p^u$ and so $(u+1)p^u < p^{2u} \leq p^{\alpha u}$. Thus this range of k does satisfy the conditions of Theorem 1.2. We consider two sub-cases: $p = 2$ and $p \geq 3$.

Firstly when $p = 2$, we consider four possibilities. When $\alpha = 2$ and $u = 2$, the only k in the range of (2) is $k = 3$. A suitable $(2, 8, 4)$ Perfect Factor over $GF(2^2)$ was presented in Example 1.1. Secondly, when $\alpha = 2$ and $u = 3$, the only k in the range of (2) is $k = 4$. A $(3, 16, 4)$ PF can be obtained by applying Lemma 4.3 to a binary $(16, 4; 3, 2)$ PM which exists by the result of [22]. Thirdly, when $\alpha = 2$ and $u \geq 4$, it is easy to verify that, for k in the range of (6.2), both $2^k > u$ and $2^{2u-k} > 2$. From the result of [22], there exists a binary $(2^k, 2^{2u-k}; u, 2)$ PM and hence by Lemma 4.3, a $(u, 2^k, 2^2)$ PF. In the fourth possibility for $p = 2$, we consider $\alpha \geq 3$. We aim to use Lemma 4.3 and we claim that there exists a binary $(2^k, 2^{u\alpha-k}; u, \alpha)$ Perfect Map for each k in the range of (2). Using the result of [22],

we need only verify that $2^k > u$—this is clear—and that $2^{u\alpha-k} > \alpha$. Now

$$\begin{aligned} u\alpha - k &> u\alpha - u - \log_2(u+1) \\ &> u(\alpha-2) \qquad \text{since } u > \log_2(u+1) \\ &\geq 2(\alpha-2) \qquad \text{since } u \geq 2 \end{aligned}$$

and, for $\alpha \geq 3$, $2^{2(\alpha-2)} > \alpha$.

Thus for $p = 2$ and k satisfying (2), we have considered all possible parameter sets.

Suppose now that $p \geq 3$. In order for there to be a suitable k in the range of (2) we need $u \geq p$. We can therefore assume $u \geq p \geq 3$ and $\alpha \geq 2$. We write $k = u + t$ where $p \leq p^t < u + 1$ and let $s = \lceil \log_p(u+1) \rceil - t$ so that

$$1 \leq s \leq \lceil \log_p(u+1) \rceil - 1$$

Clearly

$$\log_p(u - s + 1) < \log_p(u+1) \leq \lceil \log_p(u+1) \rceil = t + s. \tag{3}$$

Also for $u \geq 3$, $u + 1 < 3^{\frac{u}{2}}$, so for $u \geq p \geq 3$, $u + 1 < p^{\frac{u}{2}}$. Then $\log_p(u+1) < \frac{u}{2}$ so that $\lceil \log_p(u+1) \rceil \leq \frac{u+1}{2}$ and then $2\lceil \log_p(u+1) \rceil - 1 \leq u$. Then

$$\begin{aligned} u &\geq 2\lceil \log_p(u+1) \rceil - 1 \\ &\geq \lceil \log_p(u+1) \rceil + \lceil \log_p(u+1) \rceil - 1 \\ &\geq (t+s) + s \end{aligned}$$

so that

$$t + s \leq u - s. \tag{4}$$

Taking (3) and (4) together, we have

$$\log_p(u - s + 1) \leq t + s \leq u - s.$$

On taking the p-th powers we obtain

$$(u - s) + 1 \leq p^{t+s} \leq p^{u-s}$$

and so by the first part of this proof, there exists a $(u - s, p^{t+s}, p^{\alpha-1})$ PF, i.e. there exists a $(u - s, p^{k-(u-s)}, p^{\alpha-1})$ PF. Applying Lemma 4.4 with $\beta = 1$ to this PF we obtain a $(u - s, p^k, p^\alpha)$ PF whose sequences have linear complexities c with

$$c \leq p^k - p^{t+s}.$$

But

$$p^{t+s} = p^{\lceil \log_p(u+1) \rceil} \geq u + 1$$

and for $u \geq p \geq 3$,

$$s \leq \lceil \log_p(u+1) \rceil - 1 < u + 1.$$

Thus we can make s applications of Lemma 4.5 to our last PF to deduce that there exists a (u, p^k, p^α) PF for $u \geq p \geq 3$ and $\alpha \geq 2$.

Thirdly we consider the cases where

$$u + \log_p(u+1) \leq k \leq \alpha u.$$

We write $k = su + t$ where either $s = 1$ and $\log_p(u+1) \leq t \leq u$ or $s > 1$ and $1 \leq t \leq u$. Then $k - u = (s-1)u + t$ and

$$\log_p(u+1) \leq k - u \leq \alpha u - u = (\alpha - 1)u$$

so that by the inductive hypothesis there exists a $(u, p^{k-u}, p^{\alpha-1})$ PF. We apply Lemma 4.4 with $\beta = 1$ to obtain a (u, p^k, p^α) PF.

We have now considered all the parameter sets for PFs over an alphabet of size p^α allowed by Theorem 1.2—this completes the inductive step and with it, the proof of the theorem. □

5. Combining Perfect Factors

In this section we make use of Theorem 4.6 in a construction to combine PFs over alphabets of sizes that are powers of different primes. We obtain a large class of c-ary PFs; we discuss how far the construction of this class goes to proving the sufficiency of the conditions of Theorem 1.2 in the general case. We also give the results of recent further work in this area, [19], [20].

LEMMA 5.1 *Suppose that for $1 \leq i \leq r$, there exist (u, t_i, c_i) PFs with the c_i mutually co-prime. Then there exists a $(u, \prod_{i=1}^r t_i, \prod_{i=1}^r c_i)$ PF.*

Proof. For $1 \leq i \leq r$, let Γ_i be a (u, t_i, c_i) PF over the alphabet of integers modulo c_i.

For each choice of sequences $(s)^i = \ldots, s^i_{-1}, s^i_0, s^i_1, \ldots \in \Gamma_i$, $1 \leq i \leq r$, we produce a sequence $(w) = \ldots, w_{-1}, w_0, w_1, \ldots$ over the integers modulo $\prod_{i=1}^r c_i$ by defining w_j to be the unique solution modulo $\prod_{i=1}^r c_i$ of the equations

$$w_j \equiv s^i_j \pmod{c_i}, \qquad 1 \leq i \leq r,\ j \in \mathbf{Z}. \tag{5}$$

We claim that the set of sequences Γ' formed in this way from the original PFs is a $(u, \prod_{i=1}^r t_i, \prod_{i=1}^r c_i)$ PF.

Firstly, the sequences of Γ' each have period $\prod_{i=1}^r t_i$: a typical sequence (u) is formed from sequences $(s)^1, \ldots, (s)^r$ of co-prime periods $t_1, \ldots, t_r$ which therefore satisfy $s^i_j = s^i_{j+\prod_{i=1}^r t_i}$ for all $j \geq 0$, so that, referring to equation (5), we have $w_j = w_{j+\prod_{i=1}^r t_i}$ for all $j \in \mathbf{Z}$.

We note that there are at most $\prod_{i=1}^{r} {c_i}^u$ different u-tuples of integers modulo $\prod_{i=1}^{r} c_i$ occurring as sub-sequences in the sequences of Γ': there are $\frac{{c_i}^u}{t_i}$ sequences in Γ_i, so there are at most $\prod_{i=1}^{r} \frac{{c_i}^u}{t_i}$ inequivalent sequences of period $\prod_{i=1}^{r} t_i$ in Γ' which give rise to at most $\prod_{i=1}^{r} {c_i}^u$ different sub-sequences of length u. To show that Γ' is a PF we need only show that every possible u-tuple of integers modulo $\prod_{i=1}^{r} c_i$ does occur as a sub-sequence in some sequence of Γ'. Suppose then that $z_1, \ldots, z_u$ is a u-tuple of integers modulo $\prod_{i=1}^{r} c_i$. Define $z_j^i \equiv z_j \pmod{c_i}$ for $1 \leq j \leq u$, $1 \leq i \leq r$. Then for each $1 \leq i \leq r$, the sub-sequence $z_1^i, \ldots, z_u^i$ of length u occurs in some sequence $(s)^i$ of Γ_i, beginning say at position x_i, where $0 \leq x_i \leq t_i - 1$. Solving the system

$$x \equiv x_i \pmod{t_i}, \quad 1 \leq i \leq r,$$

uniquely for x with $0 \leq x < \prod_{i=1}^{r} t_i$ and referring to equation (5), we see that the sequence (w) of Γ' derived from $(s)^1, \ldots, (s)^r$ has $z_1, \ldots, z_u$ as the sub-sequence of length u beginning at position x. □

COROLLARY 5.2 *Suppose the integer $c \geq 2$ has prime factorisation*

$$c = \prod_{i=1}^{r} {p_i}^{\alpha_i}$$

and that t has prime factorisation

$$t = \prod_{i=1}^{r} {p_i}^{k_i}$$

with

$$u + 1 \leq {p_i}^{k_i} \leq {p_i}^{\alpha_i u}, \quad 1 \leq i \leq r.$$

Then there exists a (u, t, c) PF.

Proof. Since $u + 1 \leq {p_i}^{k_i} \leq {p_i}^{\alpha_i u}$ for $1 \leq i \leq r$, there exists a $(u, {p_i}^{k_i}, {p_i}^{\alpha_i})$ PF, Γ_i, by Theorem 4.6. The p_i's are distinct primes so we may apply Lemma 5.1 to the PFs Γ_i to obtain a PF with the desired parameters. □

Mitchell, [19], has recently strengthened Corollary 5.2 to the following result.

Result 5.3. [19] Suppose the integer $c \geq 2$ has prime factorisation

$$c = \prod_{i=1}^{r} {p_i}^{\alpha_i}$$

and that t has prime factorisation

$$t = \prod_{i=1}^{r} {p_i}^{k_i}$$

with

$${p_i}^{k_i} \leq {p_i}^{\alpha_i u}, \quad 1 \leq i \leq r$$

and

$$u + 1 \leq {p_j}^{k_j} \quad \text{for some } 1 \leq j \leq r.$$

Then there exists a (u, t, c) PF.

Thus we see that the condition that the sequences of each 'component factor' have period greater than u has been considerably weakened to the condition that there is at least one factor whose sequences have that property. In [20], Mitchell extends his methods further and establishes that, for any c and any $u \leq 4$, the conditions of Theorem 1.2 on u, t and c are in fact sufficient for the existence of a PF. However, there still remain parameter sets (u, t, c) satisfying Theorem 1.2 for which the existence of a PF is not settled, the smallest open cases appearing to be for parameter sets (5, 6, 6) and (5, 12, 6).

6. A Class of Perfect Maps

Our purpose in this section is to use Result 5.3 to produce a new class of c-ary Perfect Maps similar to the binary Perfect Maps of [6].

THEOREM 6.1 *Suppose there exists a (u, t, c) PF and v is a positive integer such that when $v = 2$ and t is even, then the condition $2t \mid c^u$ holds. Then there exists a c-ary $(t, \frac{c^{uv}}{t}; u, v)$ Perfect Map.*

Proof. Suppose there exists a (u, t, c) PF Γ; then $t \mid c^u$ and following Construction A of [6] we form a new directed graph with $\frac{c^{uv}}{t}$ vertices which are the $t \times v$ matrices whose columns are the ordered sets of sequences of Γ shifted relative to one another. The shifts we use lie between 0 and $t - 1$, and there is an edge from vertex x to vertex y in the new graph if the last $v - 1$ columns of vertex x are the same sequences as the first $v - 1$ of y and if they occur in the same relative shifts as the first $v - 1$ of y.

The method of [6] can be applied to show that there exists a Hamiltonian cycle in this new graph. Alternatively the techniques of [21] generalise easily when $c = p^\alpha$ is a power of a prime and can be used to exhibit an explicit Hamiltonian cycle in terms of p^k-ary de Bruijn and related sequences. In either case we form the $\frac{c^{uv}}{t} \times t$ array B whose consecutive columns are the first columns in the vertices of the Hamiltonian cycle. The columns of B are shifted relative to each other according to the shifting between columns of vertices of the cycle. Construction A of [6] generalised from the binary case guarantees that every possible $u \times v$ sub-array occurs in B.

To ensure that B is a Perfect Map, we need to get the last $v-1$ columns of B in the same shift as the first $v-1$ columns. Each shift occurs the same number, $\frac{c^{uv}}{t^2}$, times. The sum of all the shifts is therefore

$$\frac{c^{uv}}{t^2}(1+2+\cdots+(t-1)) = \frac{c^{uv}}{2t}(t-1)$$

Since the vertical size of the array is t, we take this sum modulo t. To get the last $v-1$ columns in the same shift as the first $v-1$ columns, we must have either $v=1$ (where there is nothing to be done) or $v \geq 2$ and

$$\frac{c^{uv}}{2t}(t-1) \equiv 0 \pmod{t}.$$

When t is odd, $2 \mid t-1$. Recalling that $t \mid c^u$ we see that this congruence holds. When t is even, $\frac{c^{uv}}{2t}(t-1) \equiv 0 \pmod{t}$ if and only if $2t^2 \mid c^{uv}$. Remembering that $t \mid c^u$ and $v \geq 2$ we see that this is the case, only when either $v \geq 3$ or $v=2$ and $2t \mid c^u$. These are the conditions on v in the statement of the Theorem. □

We now use Theorem 6.1 with the results of the previous section to obtain:

THEOREM 6.2 *Suppose the integer $c \geq 2$ has prime factorisation*

$$c = \prod_{i=1}^{r} p_i^{\alpha_i}$$

and that t has prime factorisation

$$t = \prod_{i=1}^{r} p_i^{k_i}$$

with

$$p_i^{k_i} \leq p_i^{\alpha_i u}, \quad 1 \leq i \leq r$$

and

$$u+1 \leq p_j^{k_j} \quad \text{for some } 1 \leq j \leq r.$$

Suppose further that v is a positive integer such that when $v=2$ and t is even, we have $2t \mid c^u$. Then there exists a c-ary $(t, \frac{c^{uv}}{t}; u, v)$ Perfect Map.

To date, this is the most general construction for c-ary Perfect Maps (it does not however contain the complete results of [13] and [22] as subcases, and indeed is rather far from providing a construction for Perfect Maps for every possible parameter set as can be seen from Theorem 1.3).

From a practical point of view, when c is not a power of a prime, Theorem 6.1 does not give us a good computational method for constructing Perfect Maps—for in that case there

is no directly obvious way of finding a Hamiltonian cycle in the graph of Theorem 6.1. In our final result we show how to combine PMs over alphabets of co-prime sizes in a similar way to Lemma 5.1. Coupled with the techniques of [21] generalised from the binary case we obtain a convenient method for generating Perfect Maps over c-ary alphabets.

LEMMA 6.3 *Suppose that for* $1 \leq k \leq l$, *there exist* c_k*-ary* $(r_k, s_k; u, v)$ *PMs with the* c_k *mutually co-prime. Then there exists a* $(\prod_{k=1}^{l} c_k)$*-ary* $(\prod_{k=1}^{l} r_k, \prod_{k=1}^{l} s_k; u, v)$ *PM.*

Proof. For $1 \leq k \leq l$, let $A_k = (a_{i,j}^k)$ be a $(r_k, s_k; u, v)$ PM over the alphabet of integers modulo c_k. Define an array $A = (a_{i,j})$ by defining $a_{i,j}$ to be the unique solution modulo $\prod_{k=1}^{l} c_k$ of the equations

$$a_{i,j} \equiv a_{i \pmod{r_k}, j \pmod{s_k}}^{k} \pmod{c_k},$$

for

$$1 \leq k \leq l, \quad 0 \leq i < \prod_{k=1}^{l} r_k, \quad 0 \leq j < \prod_{k=1}^{l} s_k.$$

We claim that the array A formed in this way from the original PMs is a PM with the desired parameters. The proof of this claim may be verified in a similar way to the proof of the corresponding claim of Lemma 5.1. □

We illustrate the fore-going result with:

Example 6.4. The sequences [0, 0, 1], [1, 1, 2], [2, 2, 0] form a (2, 3, 3) PF over $\mathbf{Z}_3$. Applying Theorem 6.1 we obtain the (3, 27; 2, 2) PM A_1 over $\mathbf{Z}_3$:

$$\begin{pmatrix} 001021122011122102102001220 \\ 001021122001021122001021122 \\ 112102200102001220011122102 \end{pmatrix}.$$

Let A_2 be the (4, 4; 2, 2) binary PM:

$$\begin{pmatrix} 0010 \\ 0001 \\ 0111 \\ 1011 \end{pmatrix}.$$

Applying Lemma 6.3 to A_1, A_2, we obtain a (12, 108; 2, 2) Perfect Map over $\mathbf{Z}_6$, part of which is shown below:

$$\begin{pmatrix} 00102412201442540210203422 3... \\ 00432445200102412203402142 2... \\ 41510553013503152331415543 5... \\ 30135415501112513213503152 3... \\ \vdots \end{pmatrix}.$$

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